1 Project 3: Game Theory and Behavioral Economics

Question 1.1: In the context of price wars, which prisoner's dilemma outcome is best for consumers? Why?

LOL

Question 1.4: In a tournament with n players, how many matches are run?

Hint: At an iteration i, how many players does player i play against? Do you repeat any matches?

Type your answer here, replacing this text.

Question 1.5: Consider the four strategies we've defined so far: defector, cooperator, alternator, and tit-for-tat. Which strategy do you think would win in a tournament between these four? Why?

Type your answer here, replacing this text.

In [ ]:
```
df1 = tournament_results.to_df()
df2 = tournament_results.to_df()
df2 = df2.rename({"p1": "p2", "p2": "p1", "p1_mean": "p2_mean", "p2_mean": "p1_mean"}, axis=1)
df = pd.concat([df1, df2])
df["p1"], df["p2"] = df["p1"].astype(str), df["p2"].astype(str)
df = df.pivot(index="p1", columns="p2", values="p1_mean")

plt.figure(figsize=[15,8])
sns.heatmap(df.T, annot=True, fmt=".3f")
plt.xlabel("Player 1")
plt.ylabel("Player 2")
plt.title("Player 1's Mean Scores by Opponent")
plt.xticks(rotation=90);
```

Question 1.10: Interpret the plot above. What results stand out to you? Which strategy would you choose if you were playing in a tournament? Justify your answer.

Type your answer here, replacing this text.

Question 1.11: In analyzing his tournament, Axelrod noted that one of the traits of the best strategies was being "non-envious." A strategy that is non-envious does not strive to score higher than its opponent. Why do you think this trait is linked to good strategies? How does one of the other strategies you looked at (excluding defector, cooperator, and tit-for-tat) embody this trait?

Type your answer here, replacing this text.

Question 1.12: The prisoner's dilemma is often used by economists to study and understand different phenomena that are observed in economies at different scales (e.g. studying oligopolies). Another common economic application is studying advertising and how the advertising of other firms needs to be taken into account in a single firm's advertising strategies. Describe how firms in an advertising space can be viewed as players in an iterated prisoner's dilemma. What decisions are analogous to defection and cooperation? What are the payoffs of each?

Type your answer here, replacing this text.

Question 2.4: Plot overlaid histograms of the `ErrorRatio` column grouped by `Condition`.

Hint: Recall that we can created a grouped histogram of a column of a table using `tbl.hist(..., group=GROUP_COLUMN)`.

```
In [ ]: ...
```

Question 2.5: Interpret the histogram you just created. What can you say about the distributions of the up- and down-conditions?

Hint: Look at where each histogram is centered.

Type your answer here, replacing this text.

Question 2.6: What are the null and alternative hypotheses for our A/B test?

Null hypothesis: *Type your answer here, replacing this text.*

Alternative hypothesis: *Type your answer here, replacing this text.*